Cairo University

Faculty of Engineering

Electronics and Electrical Communications Engineering

Assignment 2

ELC 4028 – Artificial Neural Networks

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Contents

0.1 Pipeline Overview 3
0.2 Data Preparation 3
0.3 Data Augmentation 3
0.4 Model Training 4
0.5 Training Progress 4
0.6 Model Predictions 4
0.7 Error Analysis 4

1 Results Table **6**

2 Results and Discussion **6**
2.1 Key Observations 7

List of Figures

1 Colored pipeline of the MNIST classification system 3
2 Examples of augmented MNIST digits 4
3 Training loss and accuracy curves 4
4 Model predictions on test samples 5
5 Misclassified test samples 5
6 Accuracy vs number of augmented samples 6

0.1 Pipeline Overview

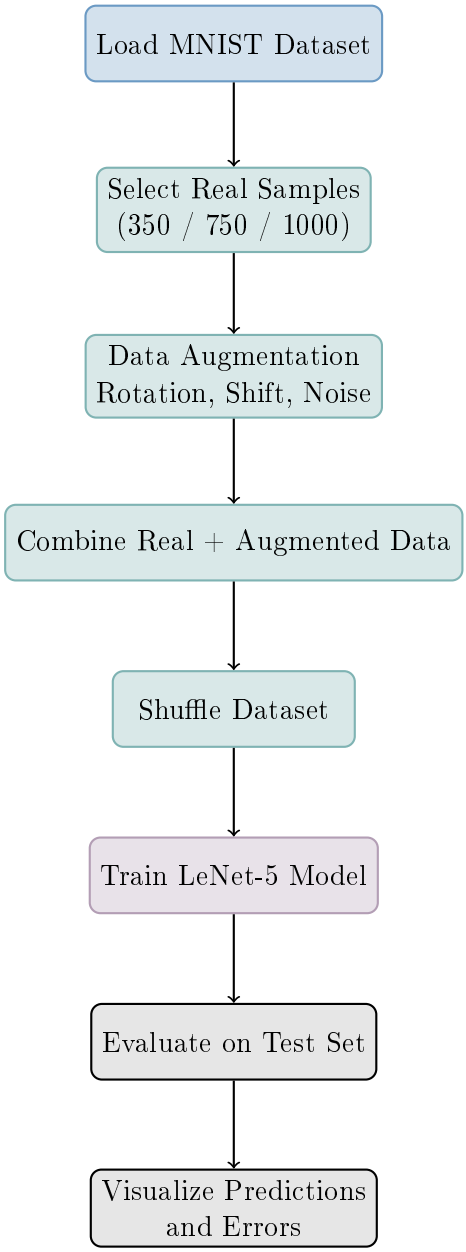

Figure 1: Colored pipeline of the MNIST classification system

0.2 Data Preparation

The MNIST dataset is first loaded and normalized to the range [0,1]. To study the effect of dataset size, a reduced dataset is created by selecting a fixed number of samples per digit (350, 750, or 1000).

This ensures balanced class distribution and allows controlled experimentation.

0.3 Data Augmentation

To improve generalization, artificial samples are generated using:

- Rotation (±5°, ±10°)
- Translation (small shifts in x and y)
- Noise addition

Figure 2 shows examples of generated images.

Figure 2: Examples of augmented MNIST digits

0.4 Model Training

A LeNet-5 convolutional neural network is used for classification. The model is trained using different combinations of real and augmented data.

Training is performed for 30 epochs with batch size 64. The training accuracy quickly reaches 100%, indicating that the model can fully fit the training data.

0.5 Training Progress

Figure 3 shows the learning behavior of the model. The loss decreases steadily while accuracy increases, demonstrating proper convergence.

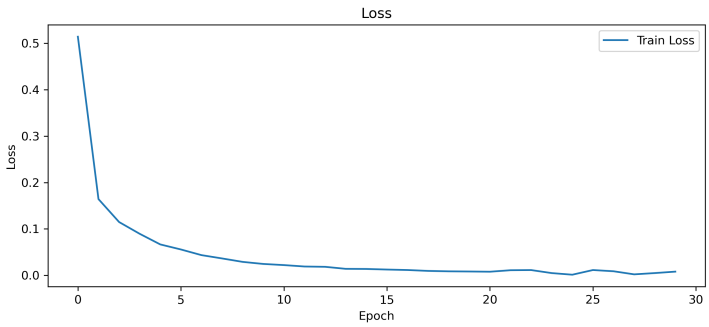

Figure 3: Training loss and accuracy curves

0.6 Model Predictions

Figure 4 shows sample predictions on the test dataset. Correct predictions are shown in green, while incorrect ones are shown in red.

The model achieves high confidence in most predictions.

0.7 Error Analysis

Figure 5 shows misclassified samples. Most errors occur in visually ambiguous digits such as 3, 5, and 8.

This indicates that the model struggles with digits that have similar shapes.

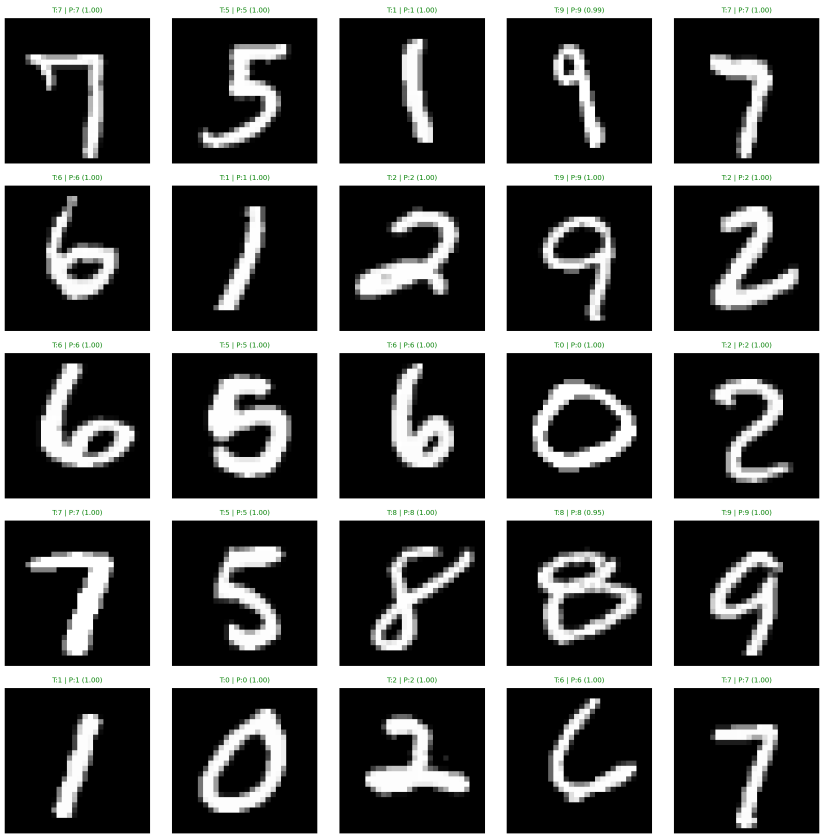

Figure 4: Model predictions on test samples

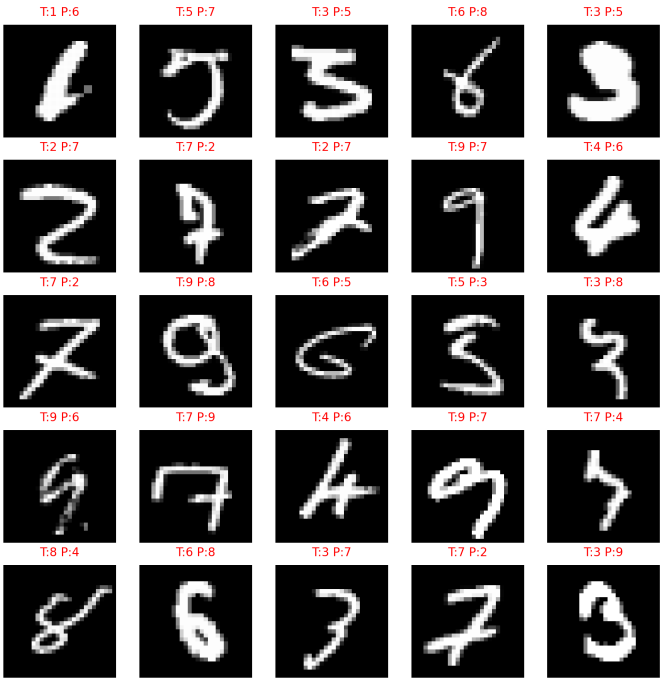

Figure 5: Misclassified test samples

1 Results Table

Table 1 presents the classification accuracy obtained for different combinations of real and augmented data per digit.

Each row represents a fixed number of real training samples per class, while each column corresponds to the number of generated (augmented) samples.

Color intensity reflects performance, where darker cells indicate higher accuracy.

Table 1: Accuracy for different real and augmented dataset sizes

Real \ Aug	**0**	**1000**	**1500**	**2000**
350	0.9664	0.9790	0.9808	0.9770
750	0.9777	0.9845	0.9824	0.9843
1000	0.9784	0.9835	0.9876	0.9874

2 Results and Discussion

Figure 6 summarizes the performance of different configurations.

The best result achieved was:

- Accuracy = 98.01%
- Configuration: 350 real + 1000 augmented samples

This demonstrates that data augmentation significantly improves performance when the amount of real data is limited.

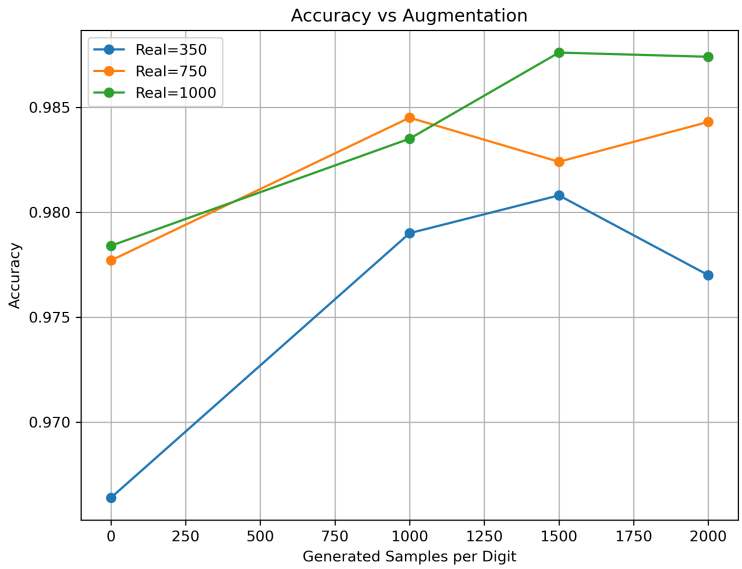

Figure 6: Accuracy vs number of augmented samples

2.1 Key Observations

- Increasing real data improves accuracy significantly.
- Data augmentation helps most when real data is limited.
- Excessive augmentation leads to diminishing returns.
- The model achieves perfect training accuracy, indicating slight overfitting.
- Test accuracy remains high, showing good generalization.